

Ege Ozkoc

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Education

University of Pittsburgh

PhD in Electrical and Computer Engineering

Aug 2025 – Expected 2028

Pittsburgh, PA

- **Research:** Machine learning, LLM-based interpretability, and signal processing for real-world time-series data, with applications to AI-ECG.
- **Cross-registered at Carnegie Mellon University:** 10-601 Introduction to Machine Learning, 10-714 Deep Learning Systems: Algorithms and Implementation
- **Selected coursework:** AI for Embedded Systems, Optimization Methods

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

MSc in Medical Engineering: Medical Image and Data Processing

Oct 2022 – Nov 2024

Erlangen, Germany

- **Master's thesis at Fraunhofer IIS:** "Evaluation of Traditional and Machine Learning Algorithms for Real-time Parkinson's Tremor Detection"
- **Selected coursework:** Deep Learning, Pattern Recognition, Image, Video and Multidimensional Signal Processing, Speech and Language Processing

Middle East Technical University

BSc in Electrical and Electronics Engineering

Aug 2016 – Aug 2022

Ankara, Türkiye

- **Capstone:** Built a wearable system for real-time hand-to-face detection and hygiene reminders during COVID-19.

Research & Industry Experience

University of Pittsburgh

PhD Researcher, Signal Processing and Statistical Learning Lab (SPSL)

Aug 2025 – Present

Pittsburgh, PA

- Developed an evidence-grounded LLM framework that combines model predictions, structured signal evidence, and attribution methods to generate interpretable explanations for AI-ECG models.
- Designed pipelines integrating deep-learning and feature-based outputs with waveform-deviation summaries, saliency maps, and SHAP attributions as structured evidence for LLM-generated explanations.
- Lead inventor on a U.S. provisional patent application for this framework; concurrently developing machine-learning models to identify inducible myocardial ischemia from resting 12-lead ECGs.

Fraunhofer Institute for Integrated Circuits IIS

Master's Thesis Researcher, Medical Sensors and Analytics Group

Mar 2024 – Nov 2024

Erlangen, Germany

- Developed an end-to-end real-time time-series classification pipeline from wrist-worn IMU accelerometer data, targeting deployment in wearable assistive systems.
- Benchmarked signal-processing, classical machine-learning, and deep-learning models including FFT-based detection, SVMs, Random Forests, CNNs, and GRUs using subject-independent cross-validation; the lightweight 1D CNN achieved an AUC of approximately 0.91 on unseen subjects.
- Optimized the CNN for resource-constrained deployment using 8-bit post-training quantization and structured and unstructured pruning; quantization reduced model size by approximately 75% with negligible performance loss, leading to a first-author publication at IEEE MLSP 2025.

FAU Erlangen-Nürnberg

Research Intern & Student Assistant, Machine Learning and Data Analytics Lab

Jun 2023 – Apr 2024

Erlangen, Germany

- Conducted MATLAB-based predictive gait simulations to study the effects of leg-length inequality in a biomechanics research project.
- Developed Python exercises for FAU's Biomedical Signal Analysis course.

Universitätsklinikum Erlangen

Student Research Assistant, Department of Radiology

Feb 2023 – Oct 2023

Erlangen, Germany

- Contributed to a neural-network project generating virtual contrast-enhanced breast MRI images.
- Built SimpleITK/Pandas workflows to preprocess MRI scans, filter patient data, and prepare model-ready inputs.

Middle East Technical University
Undergraduate Researcher, Heart Research Laboratory

Jul 2020 – Jul 2022
Ankara, Türkiye

- Applied Bayesian MAP estimation to an ill-posed inverse problem, reconstructing cardiac electrical activity from body-surface measurements for noninvasive electrocardiographic imaging.
- Built and evaluated probabilistic prior models from measured and simulated data, including MATLAB-based registration and interpolation across geometries; benchmarked Bayesian reconstruction against Tikhonov regularization.
- Demonstrated that localized prior data improved reconstruction quality and that simulated-data priors outperformed Tikhonov for sinus-rhythm reconstruction; work resulted in two peer-reviewed conference publications.

ASELSAN (*Türkiye's largest defense electronics company*)

Ankara, Türkiye

Part-time Candidate Engineer, Radar & Electronic Warfare

Nov 2021 – Jan 2022

Engineering Intern, Defense Systems Technologies

Jul 2021 – Aug 2021

- Developed MATLAB software to visualize missile trajectory and orientation from experimentally measured position and angle data; also supported RF-laboratory work.

Selected Projects

- **Open-Source Contributions to bitsandbytes:** Contributed three merged fixes to an open-source quantization library for PyTorch, including corrections for non-contiguous CPU blockwise quantization and LAMB optimizer parameter handling.
- **deep-learning-from-scratch:** Built a NumPy/SciPy neural-network framework with manually implemented forward and backward passes for fully connected, convolutional, RNN/LSTM, batch normalization, dropout, and the Adam optimizer; trained an AlexNet-style CNN on MNIST without autograd.
- **murmur:** Built a fully offline macOS dictation app using Whisper with MLX GPU acceleration, global hotkeys, and automatic text insertion; achieved approximately 3× faster transcription than CPU inference on Apple Silicon.

Skills

- **Programming & Development:** Python, C/C++, CUDA, MATLAB, Bash, Git, Linux, CMake, pybind11
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers
- **Scientific Computing & Data:** NumPy, SciPy, Pandas, Matplotlib, SimpleITK
- **LLMs & Explainable AI:** LoRA (PEFT), SHAP, saliency maps
- **Model Optimization & Deployment:** Quantization, pruning, ONNX, MLX
- **Signal Processing & Time Series:** ECG, EEG, IMU, spectral analysis, time-series modeling
- **Languages:** Turkish (native), English (fluent), German (intermediate)

Publications

- **Ege Ozkoc**, Tobias Sebastian Zech, Norman Pfeiffer, Stephan Göbl, Jürgen Frickel. *Compressed and Lightweight CNN for Real-Time Parkinson's Tremor Detection from Wearable IMU Data*. IEEE International Workshop on Machine Learning for Signal Processing (MLSP), 2025.
- **Ege Ozkoc**, Yesim Serinagaoglu Dogrusoz. *Bayesian MAP Solution of the Inverse ECG Problem with Sinus Rhythm Data: Evaluation of Simulated Training Sets*. IEEE Signal Processing and Communications Applications Conference (SIU), 2022.
- **Ege Ozkoc**, Elifnur Sunger, Kutay Ugurlu, Yesim Serinagaoglu Dogrusoz. *Prior Model Selection in Bayesian MAP Estimation-Based ECG Reconstruction*. International Conference on Measurement, 2021.

Awards & Scholarships

The Scientific and Technological Research Council of Türkiye (TÜBİTAK)
2205 – Undergraduate Scholarship Program

Oct 2020 – Jul 2022

Turkish Education Foundation (TEV)
TEV High Merit Scholarship

Dec 2017 – Jun 2021

- Selected as one of 59 new TEV High Merit Scholars of 2017.